MCSC 2201 MRP: Mini Research Problem

Ruiming Zhao

Adversarial Attacks on MNIST Image Classification

Topics:

- Computer Vision
- Neural Networks
- AI Security
- Adversarial Machine Learning

TL and DR

Motivation

Modern neural networks can achieve very high accuracy, but they may still be vulnerable to adversarial attacks.

Main Idea

Train a CNN model on the MNIST handwritten digit dataset and evaluate how different adversarial attacks affect the model.

Results

- CNN Accuracy Before Attack: 98.91%
- FGSM Attack Success Rate: 81.40%
- PGD (I-FGSM) Attack Success Rate: 99.99%
- Momentum I-FGSM Attack Success Rate: 99.86%

The results show that even highly accurate models can be fooled easily by carefully crafted perturbations.

Background

What is Image Classification?

Image classification is the task of assigning a label to an image.

Examples:

- Handwritten digit recognition
- Face recognition
- Medical image diagnosis
- Self-driving car object detection

MNIST is one of the most common benchmark datasets used to evaluate image classification models.

What Are Adversarial Attacks?

Adversarial attacks are small modifications to an input image that cause a neural network to make incorrect predictions.

Example

Original Image:

- Human sees "7"
- Model predicts "7"

Adversarial Image:

- Looks almost identical to humans
- Model predicts "1"

Why Important?

AI systems are increasingly used in:

- Healthcare
- Aviation
- Autonomous driving
- Security systems

Understanding vulnerabilities is important for building trustworthy AI.

Literature Review

FGSM (Fast Gradient Sign Method)

Introduced by:
Ian Goodfellow, is an American computer scientist and deep learning researcher best known for inventing Generative Adversarial Networks (GANs), a landmark contribution to artificial intelligence. His work has shaped modern machine learning, influencing both academic research and industrial applications in generative and adversarial modeling.

Characteristics:

- One-step attack
- Fast computation
- Easy implementation

Strengths

- Very efficient
- Low computational cost

Weaknesses

- Not always the strongest attack

PGD / I-FGSM

Characteristics:

- Multiple iterative attack steps
- Stronger than FGSM

Strengths

- Very effective
- Common benchmark attack

Weaknesses

- Slower than FGSM

Dataset and Model

Dataset

MNIST

- 60,000 training images
- 10,000 testing images
- Grayscale
- Size: 28 × 28 pixels
- 10 classes (digits 0–9)
- CNN Architecture:
  Input (1 × 28 × 28)
  ↓
  Conv2D
  ↓
  ReLU
  ↓
  MaxPool
  ↓
  Conv2D
  ↓
  ReLU
  ↓
  MaxPool
  ↓
  Fully Connected Layer
  ↓
  Output (10 Classes)

Training Configuration

Training Parameters

Optimizer:

- Adam

Learning Rate:

- 0.001

Loss Function:

- Cross Entropy Loss

Framework:

- PyTorch

Data Augmentation:

- Random Rotation
- Random Affine Transformations

Why CNN?

CNNs are especially effective for image data because convolution layers automatically learn important visual features such as edges and shapes.

Attack Methods

FGSM

Formula:

- Calculate gradient
- Move image in gradient direction
- Create adversarial example

Characteristics:

- Single-step attack
- Fastest method

PGD (I-FGSM)

Characteristics:

- Multiple gradient steps
- Stronger attack
- Higher success rate

Momentum I-FGSM

Characteristics:

- Uses momentum during optimization
- Helps avoid local minima
- Produces stronger adversarial examples

Results

Experimental Results

Attack	Recognition Rate Before Attack	Attack Success Rate
FGSM	98.91%	81.40%
PGD (I-FGSM)	98.91%	99.99%
Momentum I-FGSM	98.91%	99.86%

Analysis

FGSM successfully fooled the model most of the time.

PGD achieved nearly perfect attack success because it repeatedly optimized the perturbation.

Momentum I-FGSM also performed very well, although in this experiment PGD slightly outperformed it.